METHYLPREDNISOLONE

Pulse therapy
Slow IV infusion

CHEMICAL NAME 11β,17,21-trihydroxy-6α-methylpregna-1,4-diene-3,20-dione (sodium succinate or acetate)
CLASS Corticosteroid (intermediate-acting glucocorticoid)
TRADE NAMES Solu-Medrol (IV) · Depo-Medrol (depot) · Medrol (oral)

THERAPEUTIC USES / INDICATIONS

- High-dose IV pulse therapy: multiple sclerosis relapse, SLE nephritis, vasculitis, transplant rejection.
- Severe allergic reactions, acute severe asthma.
- Depot IM/intra-articular injection for arthritis, bursitis.
- Oral: inflammatory conditions (dose pack).

ADMINISTRATION

- **IV pulse:** infuse over at least 30 min (rapid large doses → arrhythmias, cardiac arrest).
- **IM, intra-articular** (acetate depot — never IV).
- **PO** with food in the morning.

DOSAGES

- **Pulse therapy:** 500 mg–1 g IV daily for 3–5 days.
- **Acute asthma:** 40–60 mg IV every 6–24 h.
- **Oral:** 4–48 mg/day.
- **Intra-articular:** 4–80 mg depending on joint.
- **Children:** 10–30 mg/kg/day pulse (max 1 g).

PHARMACOKINETICS

- **Onset:** IV 1 h. **Duration:** 12–36 h (depot weeks).
- Metabolised in liver.
- Excreted in urine. **t½:** 2–3 h.

PHARMACODYNAMICS

- **Glucocorticoid** ~5× hydrocortisone; minimal mineralocorticoid effect.
- Potent anti-inflammatory and immunosuppressive actions.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Hyperglycaemia, flushing.
- Insomnia, mood changes.
- Metallic taste (IV pulse).

ADVERSE EFFECTS (SERIOUS)

- Arrhythmias, cardiovascular collapse with rapid IV.
- Psychosis; severe hyperglycaemia.
- Infection; avascular necrosis.
- Long-term steroid effects; injection-site atrophy (depot).

CONTRAINDICATIONS & CAUTIONS

- Systemic fungal infection; live vaccines (immunosuppressive dose).
- Intrathecal/epidural use of depot (risk of neurological events).
- IV use of acetate suspension.

KEY INTERACTIONS

- As prednisolone (enzyme inducers, NSAIDs, antidiabetics, diuretics).

NURSING MANAGEMENT

- Infuse pulses slowly; monitor BP, HR, glucose during and after.
- Monitor mood, sleep, infection signs.
- Check correct formulation (succinate IV vs acetate depot).

PATIENT EDUCATION

- Pulse therapy may cause a metallic taste, flushing, sleeplessness.
- Report chest pain, palpitations, severe mood change, signs of infection.

KEY POINTS

- Glucocorticoid for IV pulse therapy (0.5–1 g/day).
- Depot form never IV.
- Infuse over ≥30 min.
- Hyperglycaemia, psychosis.

SOURCES CHECKED **[640]** Methylprednisolone 1 gram Powder for Solution for Injection (UK SmPC) **[447]** BTS/SIGN asthma guideline (SIGN 158) **[641]** Methylprednisolone Sodium Succinate, IM, IV (US label) **[642]** Methylprednisolone, oral (US label)
Full references p. 432–457